

Current Electricity

SHORT NOTES

Electric current

$$I \text{ (ampere)} = \frac{Q \text{ (coulomb)}}{t \text{ (second)}}$$

If I is not constant then $I = \frac{dq}{dt}$

Electric Current in a Conductor

$I = nev_d A$, where v_d = drift velocity.

Current Density

$$\vec{J} = \frac{i}{A} \hat{n} \text{ or } I = \vec{J} \cdot \vec{A}$$

Relation between J , E and V_d

In conductors drift velocity of electrons is proportional to the electric field inside the conductor as; $v_d = \mu E$

current density is given as $J = \frac{I}{A} = ne v_d = ne(\mu E) = \sigma E$

where $\sigma = ne\mu$ is called conductivity of material and we can also

write $\rho = \frac{1}{\sigma} \rightarrow$ resistivity of material.

Thus $\vec{E} = \rho \vec{J}$. It is called as differential form of Ohm's Law.

Electrical Resistance

$R \propto l$ (length of the conductor); $R \propto \frac{1}{A}$ (cross section area of

the conductor) at a given temperature $R = \rho \frac{l}{A}$ Where ρ is the

resistivity of the material of the conductor at the given temperature.

Dependence of Resistance on Temperature

If R_0 & R be the resistance of a conductor at 0°C and $\theta^\circ\text{C}$, then it is found that $R = R_0(1 + \alpha\theta)$.

Instead of resistance we use same property for resistivity as $\rho = \rho_0(1 + \alpha\theta)$.

Ohm's Law

$V = RI$. Ohm's law is applicable to only metallic conductors.

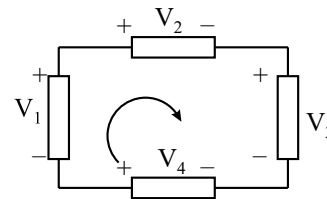
Krichhoff's Laws

1st Law (Junction law or Nodal Analysis): This law is based on law of conservation of charge.

$$\sum I_{in} = \sum I_{out}$$

It is also known as KCL (Kirchhoff's current law).

2nd Law (Loop analysis): The algebraic sum of all the voltages in closed circuit is zero. $\sum IR + \sum \text{EMF} = 0$ in a closed loop.

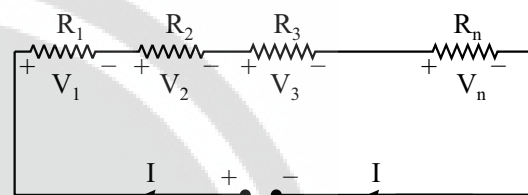


$+V_1 - V_2 - V_3 + V_4 = 0$. Boxes may contain resistor or battery or any other element (linear or non-linear).

It is also known as **KVL (Kirchhoff's voltage law)**.

Combination of Resistances

(i) Resistance in Series



$$I = I_1 = I_2 = I_3 = \dots = I_n \quad \text{and}$$

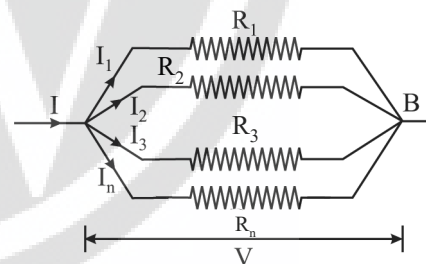
$$R = R_1 + R_2 + R_3 + \dots + R_n \quad \text{and}$$

$$V = V_1 + V_2 + V_3 + \dots + V_n$$

The voltage across a resistor is proportional to the resistance

$$V_1 = \frac{R_1}{R_1 + R_2 + \dots + R_n} V; V_2 = \frac{R_2}{R_1 + R_2 + \dots + R_n} V; \text{ etc.}$$

(ii) Resistance in Parallel



$$V = V_1 = V_2 = V_3 = \dots = V_n$$

Potential difference across each resistor is same.

$$I = I_1 + I_2 + I_3 + \dots + I_n$$

Effective resistance (R) then

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

Current in different resistors is inversely proportional to the resistances.

$$I_1 : I_2 : \dots : I_n = \frac{1}{R_1} : \frac{1}{R_2} : \frac{1}{R_3} : \dots : \frac{1}{R_n}$$

Emf of a Cell & its Internal Resistance

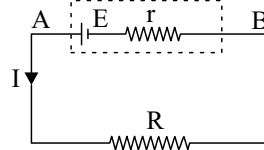
If a cell of emf E and Internal resistance r be connected with a resistance R the total resistance of the circuit is $(R + r)$.

$$I = \frac{E}{R + r}; V_{AB} = \frac{ER}{R + r}$$

where V_{AB} = Terminal voltage of the battery.

$$r = R \left(\frac{E}{V} - 1 \right)$$

If $r \rightarrow 0$, cell is ideal & $V \rightarrow E$



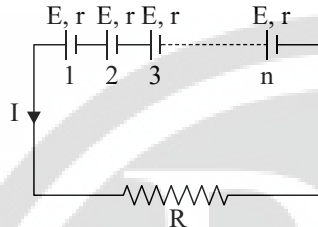
Grouping of Cells

(i) **Cells In Series:** The total emf = nE . Current in the circuit

$$I = \frac{nE}{R + nr}. \text{ If } nr \ll R$$

$$\text{then } I = \frac{nE}{R}.$$

$$\text{If } nr \gg R \text{ then } I = \frac{E}{r}.$$



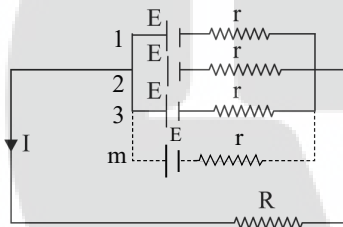
(ii) **Cells In Parallel:**

$$\text{Internal resistance of the circuit} = \frac{r}{m}.$$

$$I = \frac{E}{R + \frac{r}{m}} = \frac{mE}{mR + r}$$

$$\text{If } mR \ll r \text{ then } I = \frac{mE}{r}.$$

$$\text{If } mR \gg r \text{ then } I = \frac{E}{R}.$$



(iii) **Cells in Matrix Array:**

mn = number of identical cells.

The combination of cells is equivalent to single cell of:

(a) emf = mE &

(b) internal resistance = $\frac{mr}{n}$

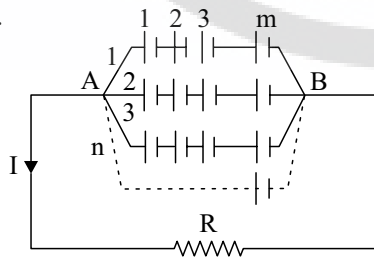
$$\text{Current } I = \frac{mrE}{mR + mr}.$$

For maximum current

$$nR = mr \text{ or } R = \frac{mr}{n}$$

$$\text{so } I_{\max} = \frac{nE}{2r} = \frac{mE}{2R}$$

For a cell to deliver maximum power across the load, internal resistance = load resistance



Wheat Stone Network

Null point or balance point $\frac{P}{Q} = \frac{R}{S}$

When,

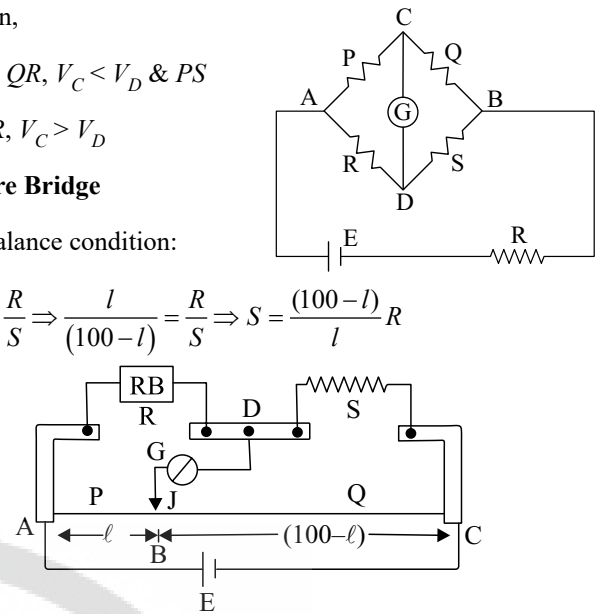
$$PS > QR, V_C < V_D \text{ \& } PS$$

$$< QR, V_C > V_D$$

Metre Bridge

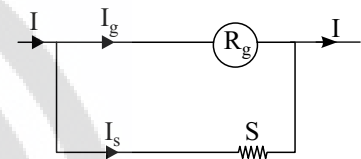
At balance condition:

$$\frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{l}{(100-l)} = \frac{R}{S} \Rightarrow S = \frac{(100-l)}{l} R$$



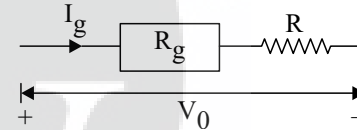
Ammeter

$$S = \frac{I_g R_g}{I - I_g}$$



An Ideal ammeter has zero resistance.

Voltmeter



$$I_g = \frac{V_0}{R_g + R}; R \rightarrow \infty, \text{ Ideal voltmeter}$$

$$R = (V/I_g) - R_g$$

Heating Effect of Electric Current

When a current is passed through a resistor, energy is wasted in overcoming the resistances of the wire. This energy is converted into heat

$$H = VIt = I^2 R t = \frac{V^2}{R} t$$

Unit of Electrical Energy Consumption

1 unit of electrical energy = kilowatt hour = 1 kWh = 3.6×10^6 joules.

❖ Series combination of Bulbs

$$\frac{1}{P_{\text{total}}} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3} + \dots$$

❖ Parallel combination of Bulbs

$$P_{\text{total}} = P_1 + P_2 + P_3 + \dots$$

RC Circuit

$$\text{❖ Charging of capacitor } q = q_{\max} \left(1 - e^{-\frac{t}{\tau}} \right)$$

$$\text{❖ Discharging of capacitor } q = q_0 e^{-t/RC}$$